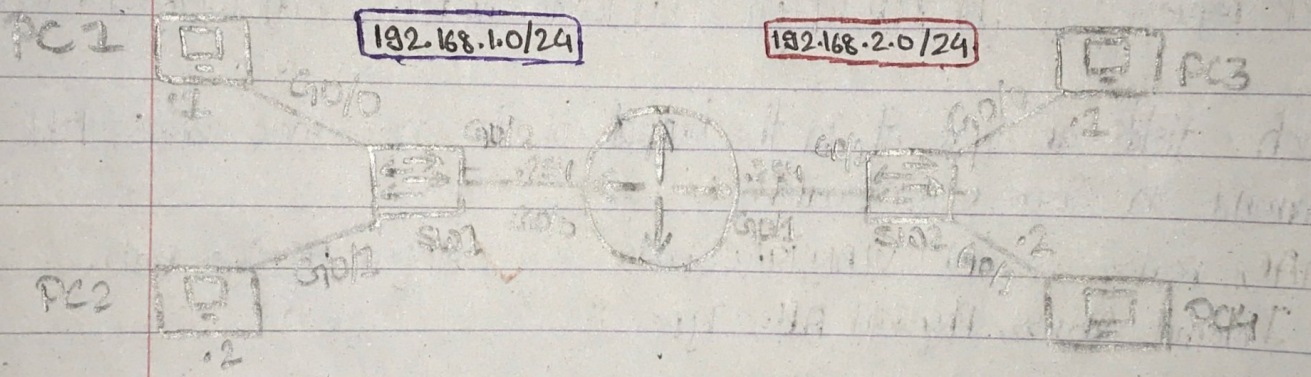
Day-7

~~Sep-6~~
Oct-6

IPv4 Addressing (Part 1)

Flashback:

- Provides connectivity between end hosts on different networks (i.e outside of the LAN).
- Provides logical addressing (IP addresses).
- Provides path selection between source & destination.
- Routers operate at level 3.



IPv4 Header.

The source & destination IP address are of 32 bit (4 byte) or 4 octet.

IP address are 32 bits (4 byte) in length.

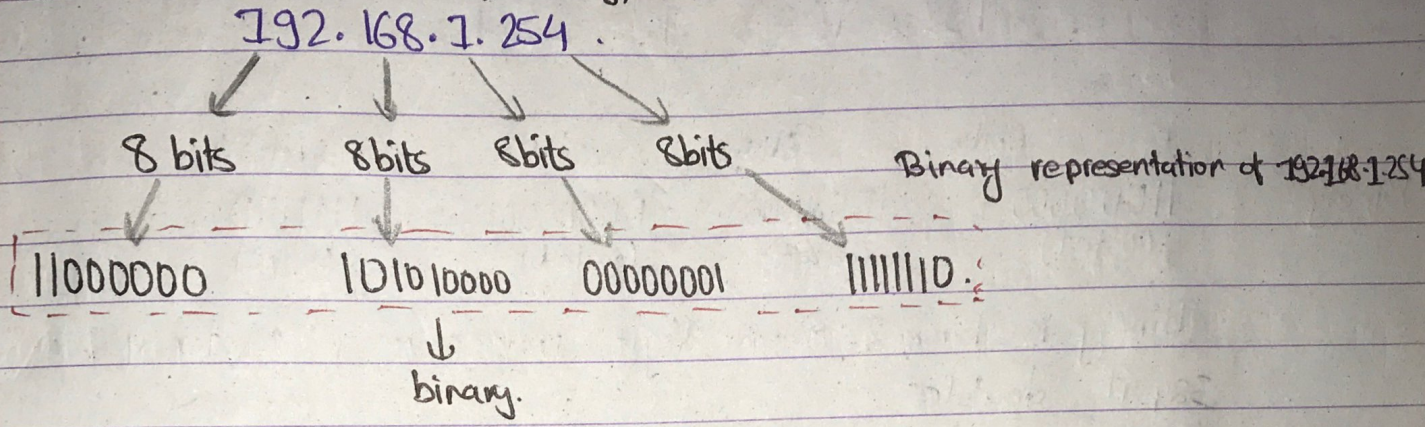
Octet	Bit
0	0
4	32
8	64
12	96
16	128
20	160
24	192

IPv4 header also includes, Time To Live, Protocol, Header Checksum, Identification etc.

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IPv4 address.

IPv4 address is 32 bits long, so each of these 4 groups of no. represent 8 bits or 1 byte.



Binary is difficult to read & remember, so IP addresses are written in dotted decimals.. There are 4 decimal nos. se each of them are separated by a dot. In above case 192 is one decimal no 168 another, 1 another & 254 another.

Decimal & Hexadecimal

Decimal. (base-10)

In decimal also known as base-10, each digit increases by a factor of 10. For eg 3294.

3	2	9	4
3×1000	2×100	9×10	4×1
3×10^3	2×10^2	9×10	4×10^0

Notice that they increase by factor of 10. Now lets look at this no. in Hexadecimal.

Hexadecimal (base-16).

Decimal 3294 is written as CDE in hexadecimal.

C	D	E.	C	D	E
$C\times256$	$D\times16$	$E\times1$	$C\times16^2$	$D\times16^1$	$E\times16^0$
(C=12)	(D=13)	(E=14)			

$3072 + 208 + 14 = 3294$

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Binary (Base-2).

192 . 168 . 1 . 254.

binary - 8bits 11000000 (octet) | 8bits 10101000 (octet) | 8bits 00000001 (octet) | 8bits 11111110 (Octet).

Binary is base-2. Each digit increase by a factor of 2 or say it doubles.

For 192 (first octet)

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0
1x128	1x64	0x32	0x16	0x8	0x4	0x2	0x1

128+64=192

For 168 (2nd octet)

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

128+32+8=168

For 1 (3rd octet)

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	1

For 254 (4th octet)

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	0

254 = 128+64+32+16+8+4+2+0

Decimal → Binary

221.

(256 > 221 > 128)

128	64	32	16	8	4	2	1
1	1	0	1	1	1	0	1

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IPv4 address.

It is a series of 32 bits. It is split into 4 octets.

11000000 10101000 00000001 11111110.

11000000

11000000 10101000 00000001 11111110.

192 . 168 . 1 . 254

192.168.1.254 /24.

Here /24 repr means that the first 24 bits (3 octets) represents the network portion of address & remaining 8 bits (1 octet) represent the end host.

So, 192.168.1 is network address & 254 is host portion

Devices on the same LAN have same Network address & their host address only differs as in our earlier eg. PC3 & PC4 have same network address 192.168.0.2 but their host address were .1 & .2 respectively.

Example

10011010 01001110 01101111 00100000

10011010 01001110 01101111 00100000

154.78.111.32 /16.

154.78 = network address

111.32 → host address

IPV4 address

Class	First Octet	First octet numeric range	Prefix length
A	0xxxxxxx	0-127	/8
B	10xxxxxx	128-191	/16
C	110xxxxx	192-223	/24.
D	1110xxxx	224-239	
E	1111xxxx	240-255.	

Class A:

If first octet begins with 0, that means the numeric range of first octet is from 0-127 as 64+32+16+8+4+2+1 = 127.

Class B:

If first octet begins with 10_2... then no. range is 128-191 as 128+32+16+8+4+1 = 191 & 128+0+0+0+0 = 128.

Class C

First octet beggining with 110, the no. range is from 192-223.

The classes of address we'll be focusing on are A, B & C. Addresses in class D are stored for multicast addresses.

We'll learn about multicast later.

Class E addresses are stored for experimental uses.

The end of class A range is usually considered to be 126, not 127. bcoz 127 range is reserved for loopback addresses.

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Loopback Addresses.

- The range is anything from 127.0.0.0 to 127.255.255.255. These addresses are used to test the 'network stack' (think OSI, TCP/IP model) on the local device.
- If a device sends any network ~~address~~ traffic to an address in this range, it is simply ~~proce~~ processed back up the TCP/IP stack as if it were ~~received~~ traffic received from another device.

```
okey@kali > ping 127.0.0.1
   Pinging 127.0.0.1 with 32 bytes of data
   Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
   . - - - -
     Packets Sent=4, Packets receive=4. loss=0(0% loss)
   Approx round trip times in milli-seconds.
   Minimum=0ms, Maximum= 0ms Average=0ms
```

Here my PC responded to its own ping, which we verified from approx round trip (min-0ms max-0ms Avg-0ms). My PC sent & received these pings to & from itself.

IPv4 addrs classes continued.

12. 128. 251. 23 /8 - Class A
154.78 .111. 22/ 16 - Class B
192. 168.1.254 / 24 - Class C.

Their prefix length match the table.

In class A there are fewer possible network address, however because the ~~network (x)host~~ host portion is very long there can be many host on each network

Class C is opposite, it has many possible network address but because host portion is smaller, there are fewer hosts on each network.

Class	Leading Bits	Size of network no field bit field	Size of rest bit field	No. of networks	Addresses per network
Class A	0	8	24	128 (2^7)	2^{24}
Class B	10	16	16	16,384 (2^{14})	65,536 2^{16}
class C	1100	24	8	2,097,152 (2^{21})	(256) 2^8

Netmask

CLASS A:	/8	255.0.0.0	11111111.00000000.00000000.00000
CLASS B:	/16	255.255.0.0	
CLASS C:	/24	255.255.255.0	

On Juniper network devices, you write prefix lengths using the (/) slash. notation. However, Cisco devices still use an older, slightly more complicated way of writing the prefix length ie using a dotted decimal netmask.

A netmask is written in dotted decimal like an IP address, where network portion is all 1's and network host portion is 0.

Network address.

If host portion of the address is all 0's, it means the it is the Network's address, the identifier of the network itself.

Host portion of the address is all 0's = Network Address.

In our e.g of first page of this lecture, we could see 192.168.1.0/24. We know that /24 means first 3 octets represent network portion & last portion represent host portion. Since the host portion is 0, it means the last octet the host portion is all 0s
∴ This address 192.168.1.0/24 is the network address.

Network address can't be assigned to host.

Network address is the first address of network, but first USABLE address is one above the network address. (in our case 192.168.1.1).

Broadcast Address.

The last address in a Network with host portion of all 1's is the broadcast address for a network. Like the network address, the broadcast address cannot be assigned to a host. Although it's the last address in the network, the last USABLE address is actually 1 under the broadcast address (in our case, 192.168.1.254)

Broadcast address is the layer 3 address used to send a packet to all hosts on the local network. If a packet is sent with this destination IP address, its MAC address would be FFFF.FFFF.FFFF.

QUIZ

Q-1) Conver the following IPv4 addr. to dotted decimal notation

00111111 00111000 11100111 00010011

↳ 63.56.231.19

Q-2) 11110011 01111111 01100010 00000001

243.127.98.1

3) 11001111 11000110 00101111 01001100

207.198.47.76

4) Convert the following IPv4 addr to binary notation.

88.46.90.91

↳ 01011000 00101110 01011010 01011011

5) 222.234.246.163

11011101 11101010 11110110 10100011